

Homework solutions for 02YELMA

Chapter: Electric Fields in Matter

1. (a) The bound volume charge density is

$$\rho_b = -\nabla \cdot \mathbf{P}.$$

In cylindrical coordinates (radial symmetry),

$$\nabla \cdot \mathbf{P} = \frac{1}{r} \frac{d}{dr}(rP_r).$$

Since $P_r = \frac{a}{r}$,

$$rP_r = a \quad \Rightarrow \quad \frac{d}{dr}(a) = 0.$$

Therefore,

$$\boxed{\rho_b = 0 \quad \text{for } 0 < r \leq R.}$$

- (b) The bound surface charge density is

$$\sigma_b = \mathbf{P} \cdot \hat{n}.$$

At $r = R$, $\hat{n} = \hat{r}$, so

$$\boxed{\sigma_b = \frac{a}{R}.$$

- (c) Total bound charge per unit length:

$$Q_{\text{vol}} = \int \rho_b dV = 0.$$

Surface contribution (per unit length):

$$Q_{\text{surf}} = \sigma_b \cdot (2\pi R) = \frac{a}{R}(2\pi R) = 2\pi a.$$

Hence,

$$\boxed{\frac{Q_{\text{total}}}{\ell} = 2\pi a.}$$

- (d) This is not contradictory: although $\rho_b = 0$ everywhere in the bulk, charge can reside on the boundary. The divergence theorem allows nonzero flux (surface charge) even if the divergence vanishes locally.
- (e) The divergence being zero only applies in the region $0 < r < R$. At the boundary, $\mathbf{P}$ terminates abruptly, producing a surface charge. Mathematically, this corresponds to a singular (delta-function-like) contribution at $r = R$.

2. (a)

$$\rho_b = -\frac{dP_x}{dx} = -\frac{d}{dx}(kx^2) = -2kx.$$

$$\boxed{\rho_b(x) = -2kx.}$$

(b) Surface charge densities:

$$\sigma_b = \mathbf{P} \cdot \hat{n}.$$

At $x = +L$, $\hat{n} = +\hat{x}$:

$$\sigma_b(+L) = kL^2.$$

At $x = -L$, $\hat{n} = -\hat{x}$:

$$\sigma_b(-L) = kL^2.$$

(c) Volume contribution per unit area:

$$Q_{\text{vol}}/A = \int_{-L}^L (-2kx) dx = 0.$$

Surface contribution:

$$Q_{\text{surf}}/A = kL^2 + kL^2 = 2kL^2.$$

Hence,

$$Q_{\text{total}}/A = 2kL^2.$$

(d) The system is *not* neutral; there is net positive bound charge.

(e) Since $\rho_b(x) = -2kx$, it is negative for $x > 0$ and positive for $x < 0$, increasing linearly in magnitude away from the center. Thus, charge density is larger in magnitude near the boundaries. Physically, this is because polarization grows with x^2 , so its spatial variation is strongest away from the center.

3. (a)

$$\rho_b = -\nabla \cdot \mathbf{P}.$$

In spherical coordinates:

$$\nabla \cdot (kr\hat{r}) = \frac{1}{r^2} \frac{d}{dr}(r^2 kr) = \frac{1}{r^2} \frac{d}{dr}(kr^3) = 3k.$$

Hence,

$$\rho_b = -3k.$$

Surface charge:

$$\sigma_b = kR.$$

(b) By symmetry, $\mathbf{E} = E(r)\hat{r}$.

Using Gauss's law inside:

$$Q_{\text{enc}} = \rho_b \frac{4}{3}\pi r^3 = (-3k) \frac{4}{3}\pi r^3 = -4\pi kr^3.$$

$$E(4\pi r^2) = \frac{Q_{\text{enc}}}{\varepsilon_0} \Rightarrow E_{\text{in}} = -\frac{k}{\varepsilon_0}r.$$

Outside:

$$Q_{\text{tot}} = (-3k) \frac{4}{3}\pi R^3 + (kR)(4\pi R^2) = 0,$$

so

$$E_{\text{out}} = 0.$$

(c) Potential at the center:

$$V(0) = \frac{1}{4\pi\epsilon_0} \left(\int \frac{\rho_b}{r'} dV' + \int \frac{\sigma_b}{R} dA' \right).$$

(d) Volume contribution:

$$V_{\text{vol}} = \frac{1}{4\pi\epsilon_0} (-3k) \int_0^R \frac{4\pi r'^2}{r'} dr' = \frac{-3k}{\epsilon_0} \int_0^R r' dr' = \frac{-3k}{\epsilon_0} \frac{R^2}{2}.$$

Surface contribution:

$$V_{\text{surf}} = \frac{1}{4\pi\epsilon_0} \frac{kR}{R} (4\pi R^2) = \frac{k}{\epsilon_0} R^2.$$

(e) Total:

$$V(0) = -\frac{3kR^2}{2\epsilon_0} + \frac{kR^2}{\epsilon_0} = \boxed{-\frac{kR^2}{2\epsilon_0}}.$$

(f) The contributions do not cancel because the spatial weighting differs: volume charge is distributed throughout the sphere with $1/r'$ weighting, while surface charge is all at $r = R$. Even though total charge is zero, the potential need not vanish.